எடுத்துக்காட்டு 11.8

கீழ்க்காண்பவற்றைத் தொகையிடுக.

(i) $5x^4$ (ii) $5x^2-4+\frac{7}{x}+\frac{2}{\sqrt{x}}$ (iii) $2\cos x\ -4\sin x\ +5\sec^2 x\ +\operatorname{cosec}^2 x$

தீர்வு

(i) $$\int 5x^4\,dx\ = 5\int x^4\,dx = 5\frac{x^{4+1}}{4+1}+c = 5\left(\frac{x^5}{5}\right)+c = x^5+c.$$

(ii) $$\int\left(5x^2-4+\frac{7}{x}+\frac{2}{\sqrt{x}}\right)dx\ = 5\int x^2dx-4\int dx+7\int\frac{1}{x}dx+2\int\frac{1}{\sqrt{x}}dx$$

$$= 5\frac{x^{2+1}}{2+1}-4x+7\log|x|+2\frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}+c$$

$$= \frac{5}{3}x^3-4x+7\log|x|+4\sqrt{x}+c$$

(iii) $$\int(2\cos x-4\sin x+5\sec^2 x+\operatorname{cosec}^2 x)dx$$

$$= 2\int\cos x\,dx-4\int\sin x\,dx+5\int\sec^2 x\,dx+\int\operatorname{cosec}^2 x\,dx$$

$$= 2\sin x+4\cos x+5\tan x-\cot x+c.$$

எடுத்துக்காட்டு 11.9

கீழ்க்காண்பவற்றைத் தொகையிடுக.

(i) $\frac{12}{(4x-5)^3}+\frac{6}{3x+2}+16e^{4x+3}$ (ii) $\frac{15}{\sqrt{5x-4}}-8\cot(4x+2)\operatorname{cosec}(4x+2)$

தீர்வு

(i) $$\int\left(\frac{12}{(4x-5)^3}+\frac{6}{3x+2}+16e^{4x+3}\right)dx$$

$$= 12\int\frac{1}{(4x-5)^3}dx+6\int\frac{1}{3x+2}dx+16\int e^{4x+3}dx$$

$$= 12\left(\frac{1}{4}\right)\left(-\frac{1}{2(4x-5)^2}\right)+6\left(\frac{1}{3}\right)\log|3x+2|+16\left(\frac{1}{4}\right)e^{4x+3}+c$$

$$= -\frac{3}{2(4x-5)^2}+2\log|3x+2|+4e^{4x+3}+c.$$

(ii) $$\int\left(\frac{15}{\sqrt{5x-4}}-8\cot(4x+2)\operatorname{cosec}(4x+2)\right)dx$$

$$= 15\int\frac{1}{\sqrt{5x-4}}dx-8\int\cot(4x+2)\operatorname{cosec}(4x+2)\,dx$$

$$= 15\left(\frac{1}{5}\right)(2\sqrt{5x-4})-8\left(\frac{1}{4}\right)(-\operatorname{cosec}(4x+2))+c$$

$$= 6\sqrt{5x-4}+2\operatorname{cosec}(4x+2)+c.$$